3 **(a)** Define *electric potential* at a point in an electric field.

.....

.....[1]

(b) Two charged spheres of radius 2.0 mm are held 4.00 cm apart (centre-to-centre). Fig. 3.1 shows the variation with displacement x, of the electric potential V, between the two charged spheres. The charge on each sphere may be considered to be concentrated at its centre.

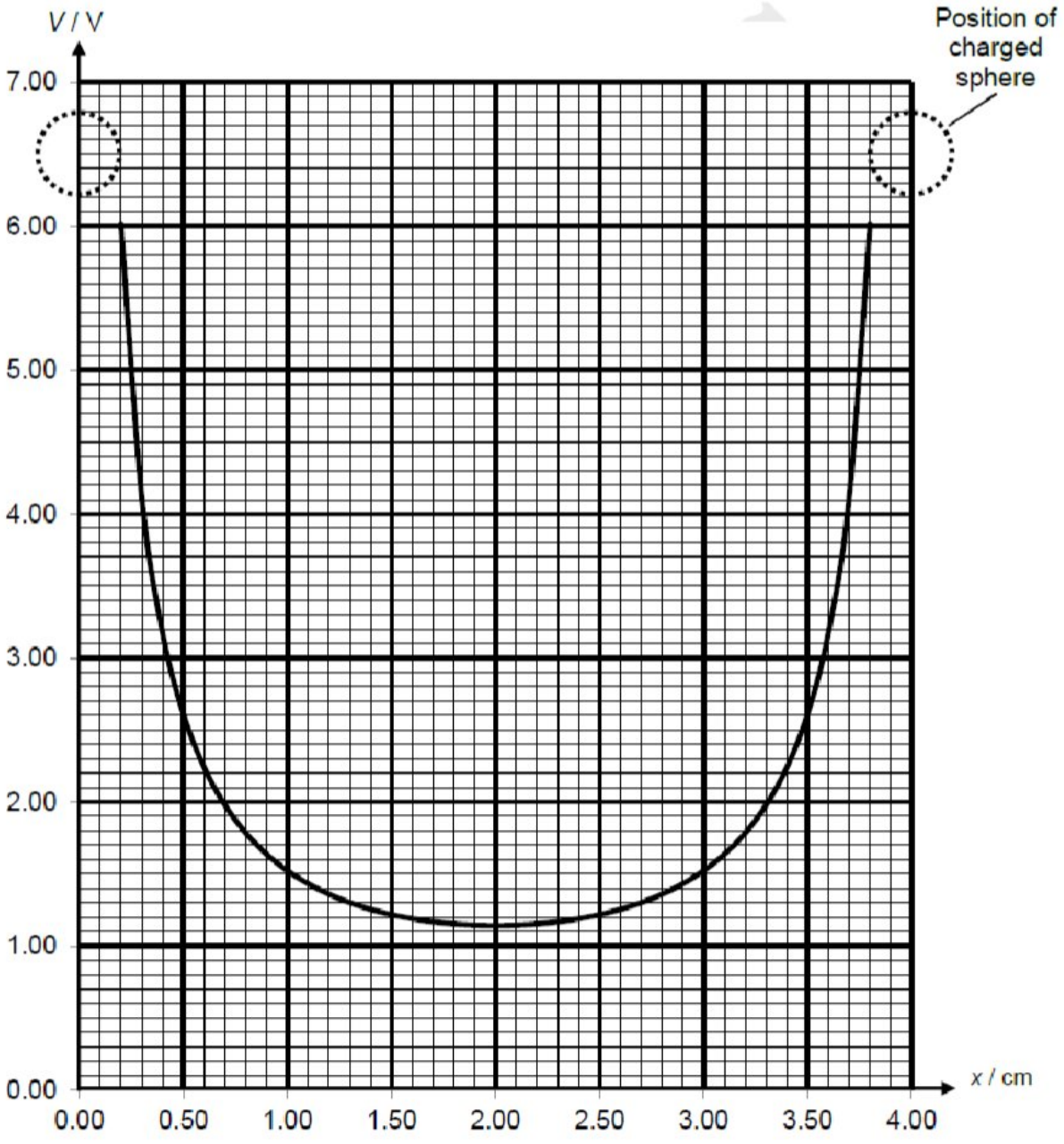


Fig. 3.1

(i) Using Fig. 3.1, determine the magnitude of the electric field strength at x = 1.00 cm and state its direction.

electric field strength = ... N C^{-1} [2]

direction: ..... [1]

(ii) Explain, using Fig. 3.1, how it can be deduced that the charges on the two spheres are equal.

.....

.....[1]

(iii) Using Fig. 3.1, determine the charge on each sphere.

charge = ... C [2]

[Total: 7]